

Case Study

SaaS Subscription & Churn

Problem Statement

The SaaS company is seeing revenue growth stall despite rising signups. Leadership suspects churn is the root cause but doesn't know which customer segments are most at risk or what's driving them to cancel.

Data Overview

2,000 customer subscription records covering plan type, acquisition source, signup and cancellation dates, MRR, and customer lifetime value across a 2-year period (2022–2023).

Questions:

1. What's the overall monthly churn rate?
2. Which subscription plan has the highest churn?
3. What's the average customer lifetime (in months) before cancellation?
4. Is churn higher for customers who signed up via a discount/trial vs. full price?
5. What's the Monthly Recurring Revenue (MRR) trend over the last 12 months?
6. Which customer segment (by plan or signup cohort) has the best retention?

Data Cleaning (Excel)

- Verified no duplicate **CustomerIDs** across 2,000 records
- Formatted **SignupDate** and **CancelDate** columns as proper date types — **CancelDate** intentionally blank for active customers, not a data error
- Added **ChurnFlag** column ($=IF(Status="Churned",1,0)$) and **TotalRevenue** column for pivot analysis
- Pivot check confirmed plan distribution: ~50% Basic, ~35% Pro, ~15% Enterprise

Analysis (SQL)

Ran 6 queries covering: overall churn rate, churn by plan, average tenure and LTV by plan, discount vs full-price churn comparison, MRR trend, and retention by acquisition source.

Key Findings

1. What's the overall monthly churn rate?

Overall churn rate is **29.55%** — meaning roughly 1 in **591** customers cancel each month.

Total_Customers	Churn_Customer	Churn_Rate_Pct
2000	591	29.55

2. Which subscription plan has the highest churn?

Basic plan has the highest churn at **30.92%** vs Enterprise at **8.68%** — price sensitivity and lower perceived value are likely drivers.

Plan	Total_Customers	Churn_Customer	Churn_Rate_Pct	Avg_MRR
Basic	1002	390	38.92	29058
Pro	710	176	24.79	56090
Enterprise	288	25	8.68	57312

3. What's the average customer lifetime (in months) before cancellation?

Average customer lifetime before cancellation is only **9.21 months** on Basic vs **10.28 months** on Enterprise — a major LTV gap. Customers across all plans cancel within a similar window — between 9.2 and 10.3 months — but the revenue impact of losing an Enterprise customer (₹2,045) is 7.6x higher than losing a Basic customer (₹267). This means churn is not just a volume problem — it's a value problem

Plan	Status	Avg_Tenure_Month	Avg_LTV
Basic	Active	18.10	524.89
Basic	Churned	9.21	267.17
Enterprise	Active	18.50	3681.12
Enterprise	Churned	10.28	2045.72
Pro	Active	18.33	1447.74
Pro	Churned	9.76	770.70

4. Is churn higher for customers who signed up via a discount/trial vs. full price?

Customers acquired via **Free Trial/Discount** churn at **27.27%** vs **29.96%** for **Full-Price** — It suggests the trial experience is doing a good job of attracting genuinely interested customers — the bigger retention problem lies with full-price customers whose expectations may not be met after paying full price from day one.

Customer_Type	Total_Customers	Churn	Churn_Rate_Pct	Avg_LTV
Full Price	1692	507	29.96	1159.90
Discount/Trial	308	84	27.27	1269.14

5. What's the Monthly Recurring Revenue (MRR) trend over the last 12 months?

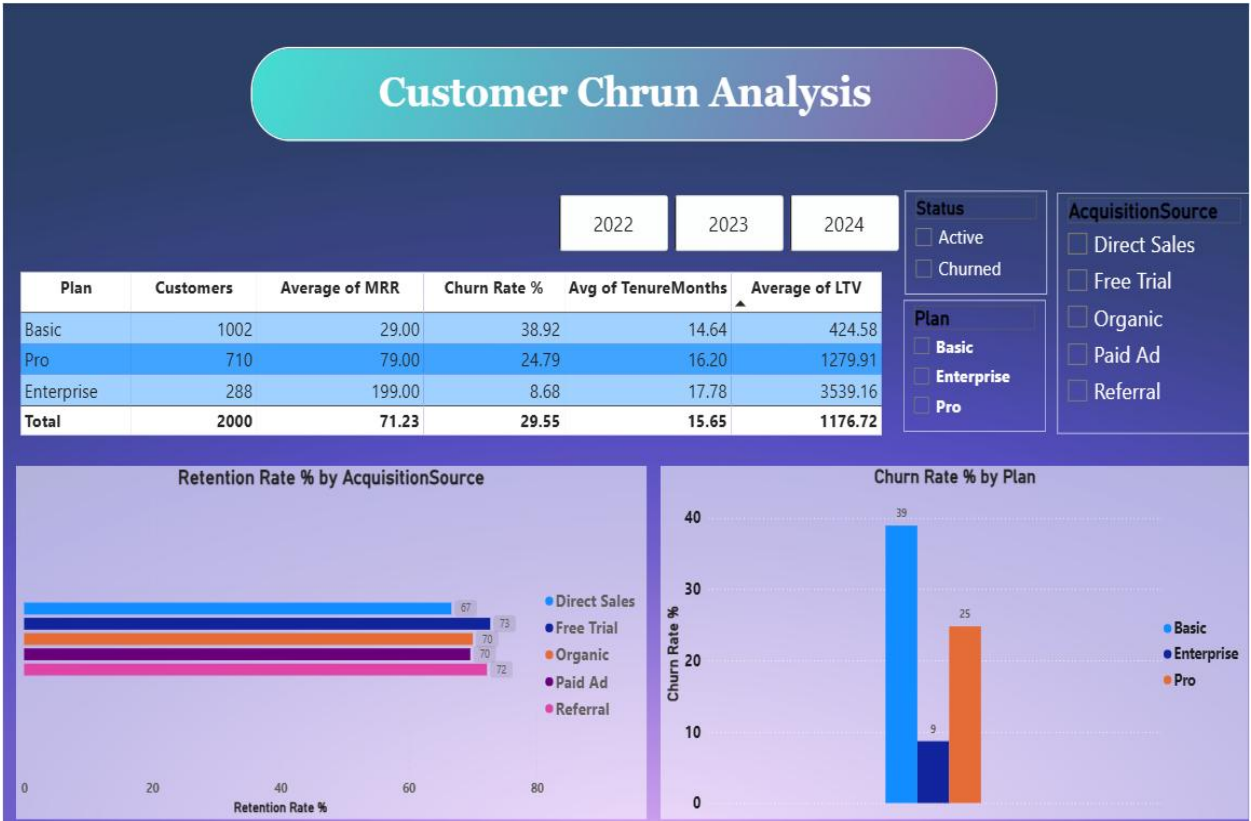
Direct Sales acquisition channel has the best retention rate at **33.33%** and highest average LTV — worth increasing investment here.

AcquisitionSource	Total_Customers	Active_Customer	Retention_Rate_Pct	Avg_LTV
Direct Sales	186	124	33.33	1128.43
Referral	396	286	27.78	1200.65
Organic	590	413	30.00	1110.41
Free Trial	308	224	27.27	1269.14
Paid Ad	520	362	30.38	1196.28

6. What's the Monthly Recurring Revenue (MRR) trend over the last 12 months?

MRR has been growing steadily since **January 2022**, reaching a cumulative MRR of **142,460** by **January 2024**. Monthly new MRR generally increased over time, with particularly strong months such as **January 2023 (7,606 MRR)** and **December 2023 (7,246 MRR)**. However, the sharp drop in **January 2024 (199 MRR)** appears to represent incomplete monthly data rather than an actual decline.

Dashboard (Power BI)



Recommendations.

1. **Focus on the Basic plan first.** It has the highest churn rate (30.92%) and also has the most customers. Improving this plan can have the biggest impact on revenue. Add a simple onboarding guide and send a follow-up email after 30 days to help new customers get started.
2. **Encourage Basic users to upgrade to Pro.** Offer a limited-time discount or extra features within the first 60 days. This can help customers see more value and stay longer.
3. **Invest more in the Direct Sales channel.** Customers from this channel stay longer and spend more over time. Bringing in more customers this way can improve long-term revenue.
4. **Find customers before they leave.** If a Basic plan customer hasn't logged in for 14 days, send them a reminder email or offer help. A small action at the right time can prevent cancellations.
5. **Offer an annual subscription plan.** Paying for a full year can encourage customers to stay longer and reduce monthly cancellations while giving the business more predictable revenue.

Tool stack:

Excel (cleaning) → SQL (analysis) → Power BI (dashboard)